A Knowledge Graph-Based RAG-GLLM Approach for Literature Review and Knowledge Discovery

Research-in-progress

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Abstract

The rapid growth of academic publications has intensified the challenges of conducting rigorous literature reviews, pushing traditional and computational approaches to their limits. While Generative Large Language Models (GLLMs) offer advanced capabilities for summarisation, abstraction, and "synthesis", their use in scholarly contexts is constrained by hallucination, limited context windows, and weak evidential grounding. Retrieval-Augmented Generation (RAG) mitigates some of these issues by anchoring outputs in verifiable sources, yet current applications remain focused on fact retrieval rather than higher-level conceptual integration. To address this gap, we propose a knowledge graph-based RAG–GLLM approach that combines structured knowledge representation with generative reasoning. Knowledge graphs provide persistent context, reduce hallucination, and enable analysis across five levels of literature review, from concepts to themes. Our research-in-progress outlines the conceptual design, development plan, and early prototyping results, contributing a methodological foundation for scalable, reliable, and theory-informed literature reviews in Information Systems research.

Keywords

Literature Review, Generative AI, Retrieval-Augmented Generation, Knowledge Graphs, Large Language Models

1 Toward Next-Generation Literature Reviews

Literature review is a foundational activity in the scientific process. It consolidates existing knowledge, situates new research within established streams, and identifies emerging opportunities for theoretical and practical advancement (Webster and Watson 2002). However, the exponential growth in academic publications, driven by improved research infrastructure, global collaboration, and increasingly powerful research tools, has fundamentally altered both the scale and complexity of this task. Traditional literature review approaches, whether narrative (Baumeister and Leary 1997), systematic (e.g., Recker et al. 2019), or bibliometric (Donthu et al. 2021), were developed in an era when publication volumes were more manageable and analytical tools were primarily manual. These methods now face increasing strain, often forcing researchers to choose between breadth of coverage and depth of synthesis. The challenge is no longer simply to summarize existing work, but also to produce theoretically informed integrations that connect disparate findings, identify gaps, and generate new avenues for inquiry.

Over the past two decades, computational text analysis methods such as Latent Semantic Analysis (LSA) and Latent Dirichlet Allocation (LDA) have been used to partially automate literature mapping and topic identification (Indulska et al. 2012; Lee et al. 2010). These approaches detect statistical co-occurrence patterns and organise terms into latent topics, offering scalable ways to summarise large corpora. However, they are fundamentally restricted to surface-level statistical associations. Because they lack a deep semantic understanding of context, their outputs often require extensive human interpretation to connect identified patterns to theoretical constructs or to integrate across heterogeneous streams of literature (e.g., Sidorova et al. 2008). This "last mile" problem, transforming statistical patterns into theoretically informed synthesis, remains largely manual. These methods may help address the challenge of scale, but are limited in conceptual depth in analysing literature.

Recent advances in natural language processing have introduced Generative Large Language Models (GLLMs) such as ChatGPT, which go beyond statistical topic modelling by leveraging deep contextual embeddings and vast pre-trained knowledge (de Kok 2025). In addition to processing and generating natural language with remarkable fluency, GLLMs can integrate semantic understanding with advanced analytical tasks such as summarisation, concept extraction, and relationship mapping. Crucially, they can detect latent semantic patterns that extend beyond surface-level keyword or co-occurrence analysis, enabling more conceptually rich and theoretically relevant interpretations. In this capacity, GLLMs offer a more powerful and flexible alternative to traditional computational methods; one that can move literature reviews beyond descriptive topic aggregation toward richer, integrated, and theoretically informed synthesis[1] (e.g., Tuczek et al. 2025).

Despite their potential, the use of GLLMs for literature review is constrained by several major limitations. Chief among these is hallucination, the tendency of models to generate plausible but factually incorrect statements (Wasserkrug et al. 2024). This is especially problematic in scholarly contexts, where traceable evidence and logical rigour are essential. Additional limitations include limited context windows for analysis (An et al. 2024), which restricts GLLMs' ability to process extensive bodies of literature as a whole, and the absence of robust evaluation metrics for assessing both the factual accuracy and the theoretical coherence of generated analyses. Consequently, although GLLMs can produce fluent and seemingly insightful outputs, their reliability as stand-alone tools for rigorous academic synthesis remains debatable.

A promising response to these challenges is Retrieval-Augmented Generation (RAG), which grounds generative outputs in verifiable external source material (Arslan et al. 2024; Wasserkrug et al. 2024). By coupling GLLMs with retrieval mechanisms anchored in trusted and verifiable academic databases, RAG mitigates hallucination and enhances transparency. For example, the RAG-powered system PaperQA retrieves and evaluates relevant paragraphs from full-text scientific articles and uses them as grounded context to inform answer generation, ensuring that outputs remain traceable to the original literature and improving factual reliability in scientific question-answering tasks (Lála et al. 2023). While RAG improves factual reliability, it does not capture conceptual relationships, as it relies on retrieving similar text embeddings, and the results are often fragmented and lack integration.

To overcome this limitation and promote the responsible use of GLLMs in literature review (and, ultimately, in scientific knowledge discovery), we propose a knowledge graph-based RAG-GLLM

[1] Some may argue that synthesis is a high-level human cognitive skill, while GLLMs, as machine learning models, operate by detecting patterns in training data to make predictions. From this perspective, GLLMs cannot truly "synthesize" knowledge in the human sense. We do not attempt to resolve this conceptual debate in this paper. For our purposes, we treat GLLMs' capability to identify, surface, and summarize latent semantic relationships as a form of "synthesis."

approach for literature review and knowledge discovery (see Figure 1). Knowledge graphs provide structured, machine-readable representations of concepts and their interrelationships (Watson et al. 2024), enabling GLLMs to maintain broader contextual awareness and reduce hallucination. [2] Furthermore, the analysis of individual articles can be recorded in the knowledge graph and reused in the analysis of subsequent articles, thereby mitigating context window limitations. In the next sections, we review the relevant literature to position the novelty of our methodological approach, outline the conceptual foundation of the design, and present our research plan along with early findings from initial prototype development. This research primarily advances literature review methodological approaches and, when applied within knowledge domains, holds potential to contribute to theoretical development.

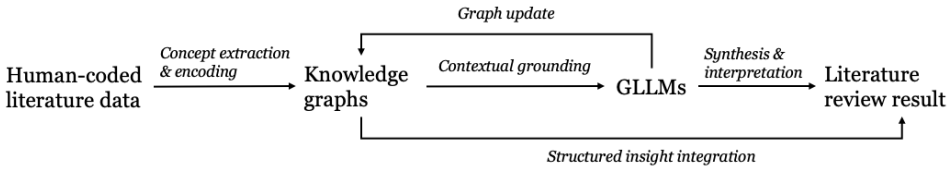


Figure 1. The High-level Overview of the RAG–GLLM Literature Review Approach

2 The Literature

2.1 Generative Large Language Models for Literature Review

Generative Large Language Models (GLLMs) represent an evolution of large language models (LLMs) that combine deep contextual understanding with the ability to produce novel textual outputs (de Kok 2025). The generative nature of GLLMs allows researchers to explore new modes of knowledge discovery and hypothesis generation. By integrating retrieved or structured information into coherent outputs, GLLMs go beyond simple pattern recognition or statistical summarisation. In research contexts, this positions GLLMs as a promising tool for tasks requiring both semantic comprehension and creative synthesis, such as semantic analysis (e.g., Yoganarasimhan and Iakovetskaia 2024), scenario planning (e.g., Chang et al. 2024), and hypothesis development (e.g., Zhou et al. 2024).

In the context of literature review, GLLMs have been applied across multiple stages. In the literature search and retrieval process, GLLMs can understand research queries, expand search terms semantically, and retrieve more comprehensive and relevant datasets (Bernard et al. 2025). During screening and selection, GLLMs have been used to automate laborious filtering tasks, interpreting human-defined inclusion or exclusion criteria and labelling articles as “accept” or “reject” (Dennstädt et al. 2024). After filtering the literature, GLLMs can help generate summaries or structured data about the literature dataset, for example, summarising work in themes and topics (e.g., ChatGPT Deep Research) or extracting information based on human queries (Tang et al. 2024). Further, GLLMs have been explored for knowledge discovery, such as defining concepts (Tuczek et al. 2025), mapping conceptual relationships (Taleb et al. 2024), and generating research hypotheses (Tong et al. 2024).

Despite these promising applications, evidence highlights limitations in reliability and factual accuracy. A key challenge is hallucination, where models generate plausible but incorrect information. For example, Delgado-Chaves et al. (2025) reported that GLLMs misclassified up to 70% of papers compared to human reviewers. Similarly, Elicit, a GLLM-based system for systematic literature review, was found to underperform in extracting information from 50% of sampled studies relative to human evaluation (Bianchi et al. 2025). These shortcomings often arise from the probabilistic nature of GLLM generation and constraints such as limited context windows, leading to inaccurate information.

Given that literature review and knowledge discovery constitute the foundational core of scientific research, requiring high levels of accuracy, coherence, and reliability, addressing the hallucination problem is essential. Only by mitigating these limitations can the creative potential of GLLMs be responsibly harnessed to support rigorous, theory-informed literature review.

2.2 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a framework that combines GLLMs with external knowledge retrieval to overcome its intrinsic limitations. GLLMs can generate fluent text but often “hallucinate” facts or rely on outdated training data. RAG addresses this by retrieving relevant external information (e.g., from scientific papers or databases) and appending it to the GLLM’s context, thereby grounding

[2] Knowledge graph is a pivotal technology in natural language processing, which is the foundation for GLLMs.

the generation in up-to-date, factual information. In practice, a user query is used to search an external document corpus (retrieval), the top results are concatenated with the GLLM query (augmentation), and then the GLLM generates an answer based on this extended prompt (generation) (Wasserkrug et al. 2024). This process reduces the model's reliance on parametric memory and significantly enhances factual accuracy and relevance of responses, especially in knowledge-intensive domains.

Beyond its technical refinements (Gao et al. 2023; Huang and Huang 2024), previous studies have highlighted RAG's versatility across domains such as medicine, finance, technology, and communication (Arslan et al. 2024). Applications span tasks from question answering and information retrieval to text analysis, report generation, and decision support. Within academic research, RAG has been explored in literature-related tasks, including answering scholarly questions (Lála et al. 2023), summarising articles (Agarwal et al. 2024), and grounding citations in generated text (Ye et al. 2023). These applications illustrate RAG's promise but also expose a gap: most reviewed existing systems remain tailored to fact-checking or descriptive summarisation, rather than supporting higher-level conceptual synthesis that is essential for theory-building in Information Systems and related fields. For instance, Lála et al. (2023) demonstrate a tool that answers specific factual queries (e.g., identifying which molecules are associated with certain macrophage responses), but such fact retrieval falls short of enabling integrated, theory-driven knowledge development.

A further limitation lies in the types of knowledge sources used. Current RAG implementations typically rely on broad, publicly available corpora, such as Wikipedia, PubMed abstracts, or Amazon book reviews, that are not optimized for scholarly synthesis (Arslan et al. 2024). While these sources provide breadth, they lack the structured depth, domain specificity, and theoretical framing required for academic literature reviews. Moreover, most approaches treat retrieval as a one-off process, appending raw documents to prompts without structuring or iteratively refining them. Few studies have explored the design of discipline-specific retrieval corpora, nor have they experimented with looped feedback architectures where GLLM outputs are fed back into the retrieval pipeline for progressive knowledge refinement. For literature review and knowledge discovery, where accuracy, conceptual coherence, and theoretical integration are paramount, these omissions represent critical research challenges.

3 Conceptual Foundation

3.1 Literature Review Tasks and Their Challenges

Scientific literature is not a flat collection of texts but a structured system where knowledge is produced, encoded, and debated at multiple levels of abstraction (Watson et al. 2024). For a literature review to be rigorous and scalable, it must operate across these layers rather than treating publications as undifferentiated sources. We conceptualize the review process as spanning five analytical levels—concept, relationship, model, theory, and theme—each with distinct goals and challenges (see a summary in Table 1).

Level	Key tasks	Key challenges
Concept	Extract definitions and identify measurement approaches for key constructs	Overlapping terms (synonymy), same labels with different meanings (polysemy) (Larsen and Bong 2016), evolving terminology across studies
Relationship	Identify reported associations, causal claims, and conditional effects	Distinguishing explicit vs. implicit links, capturing methodological form (causal vs. correlational) (Jeyaraj and Dwivedi 2020), accounting for contextual qualifiers (Avgerou 2019)
Model	Reconstruct frameworks linking multiple concepts and relationships	Fragmented or incomplete reporting, inconsistent diagram formats, heterogeneity in study designs (Jeyaraj and Dwivedi 2020)
Theory	Map how theories are invoked, extended, or contested across studies	Conflicting epistemic assumptions, varying levels of abstraction, selective or inconsistent use of theory (Gregor 2006)
Theme	Synthesize patterns into broader narratives about field evolution	Scope ambiguity (where a theme begins/ends), interpretive subjectivity, integrating across time and subdomains (e.g., Sidorova et al. 2008)

Table 1. Analytical Levels in Literature Review and Their Associated Challenges

3.2 GLLM Capabilities and Their Strength for Literature Review

GLLMs possess a set of fundamental capabilities that, when combined with retrieval, determine their utility for rigorous literature review. For clarity and parsimony, we emphasise three core capabilities:

(1) textual extraction (Xu et al. 2024), (2) semantic abstraction (Abu-Salih et al. 2025), and (3) knowledge synthesis (Babaei Giglou et al. 2024). This section shows how each core capability supports the five analytical levels and then describes the methodological limitations that our RAG–GLLM design seeks to overcome (a summary is provided in Table 2). The order reflects how reviewers typically work: first they locate and collect evidence inside and across papers, then they normalise and compress those fragments, and finally they assemble them into cross-paper explanations and narratives.

Capability	Application across review levels	Key limitations	Design objectives
Textual extraction	Identifies concepts, relationships, models, theories, and themes within source papers; retrieves definitional phrases, methodological details, and measurement descriptions.	Extraction may conflate frequency with validity; lacks explicit provenance; limited by context window and chunking granularity.	Enforce token-level provenance; apply adaptive chunking to improve logical segmentation and retrieval precision.
Semantic abstraction	Normalises varied expressions across papers; enables comparison of models and theories stated in heterogeneous terminology.	Risk of hallucination in paraphrasing; may overgeneralise; context loss without explicit grounding.	Engineer prompt templates; constrain generation to retrieved information linked via knowledge graph nodes.
Knowledge synthesis	Integrates evidence across documents to construct higher-order models, reconcile conflicting findings, and propose theoretical framings or research gaps.	Fragile reasoning if retrieval is incomplete; prone to premature convergence; lacks uncertainty calibration.	Implement iterative retrieval–generation cycles; design human-in-the-loop checkpoints and activities.

Table 2. Mapping GLLM Capabilities to Literature Review Levels

Textual extraction is the most basic capability of GLLMs: it locates relevant spans such as definitions, measures, associations, or theoretical framings within papers. Applied consistently across the five review levels, extraction answers the fundamental question: "*What does this paper say about x?*" However, similarity-based retrieval alone is insufficient; returned spans must be explicit and verifiable to ensure scholarly rigour.

Semantic abstraction builds on extraction by normalising and compressing evidence so it can be compared across studies. This enables reviewers to align varied definitions, distinguish causal from correlational claims, or summarise competing models and theories. At higher levels, abstraction supports concise narratives about field development. Yet, it carries risks: paraphrasing may smooth over disagreements or introduce unsupported links. To remain reliable, abstractions must always be annotated with references to their source text.

Knowledge synthesis integrates extracted and abstracted evidence across documents into higher-order outputs such as reconstructed models, theory maps, and thematic narratives. While synthesis is powerful, it is fragile—overly dependent on representative evidence and deliberate counterevidence checks. Without these safeguards, integrative claims risk amplifying bias or overstating consensus.

Despite these capabilities, conventional GLLMs face recurring limitations: hallucination, loss of context through chunking, lack of uncertainty signals, and weak provenance. These issues can yield persuasive but untrustworthy syntheses that fall short of academic standards. Our design therefore emphasises (1) token-level provenance, (2) adaptive chunking to preserve context, (3) evidence-constrained prompt engineering, (4) iterative retrieval–generation cycles that build persistent knowledge graphs, and (5) human-in-the-loop interventions to ensure reliability.

4 Research design

We propose a staged, multi-level process to develop and evaluate our RAG–GLLM approach. This process builds directly on the three core capabilities of GLLMs and applies them across the five levels of literature analysis. At each level, research activities will be organised into two stages: development (architecture, prompts, and workflows) and evaluation (task benchmarks and expert assessment).

Following the high-level design of the RAG–GLLM pipeline (Figure 1), the developmental tasks include: (1) constructing a knowledge graph tailored for literature review, with nodes representing extracted entities (concepts, relationships, models, theories, themes) and edges capturing their relationships; (2) designing the RAG process so that GLLMs can reason over the knowledge graph while maintaining access to source-level evidence; (3) establishing mechanisms for verification and feedback loops, where generated outputs are checked and inserted back to the knowledge graph (with human inventions); and (4) creating a final synthesis process to generate literature review outputs.

In the evaluation stage, we will empirically assess the RAG–GLLM approach through comparative experimentation. Specifically, we will benchmark it against (1) a baseline using simple, non-retrieval GLLM prompting, and (2) traditional manual literature review methods. These evaluations will focus on three dimensions of performance: efficiency (time and resource requirements) (Vindrola-Padros and Johnson 2020), accuracy (fidelity of extracted and synthesised knowledge) (Levitt et al. 2021), and creativity (ability to surface novel connections or research gaps) (Amabile et al. 1996). Both quantitative measures (e.g., precision, recall, F1, inter-rater agreement) and qualitative assessments (e.g., expert evaluation of review outputs) will be employed.

We are currently developing the first prototype of our RAG–GLLM literature review approach, with a particular focus on concept extraction. This stage is foundational because accurate identification of key constructs underpins the subsequent relationship, model, theory, and theme analyses. Our initial development has concentrated on the first two core development tasks (see Figure 2). To evaluate its feasibility, we conducted a pilot test using one research article (Rai et al. 2009). Results indicate that our approach substantially outperforms a non-retrieval GLLM baseline (see Table 3). While the baseline produced a large set of loosely related concepts with low precision and recall, the RAG–GLLM method retrieved nearly all core constructs present in the study's theoretical model with markedly higher accuracy (Precision = 43%, Recall = 93%, F1 = 59%). Some noise remains, attributable to the use of an intentionally simple prompt. We are now refining prompt engineering following de Kok (2025) and Wasserkrug et al. (2024) to improve both precision and contextual relevance. These early findings demonstrate that the integration of knowledge graphs with retrieval can significantly enhance the reliability of concept extraction, thereby highlighting the utility of our proposed RAG-GLLM approach.

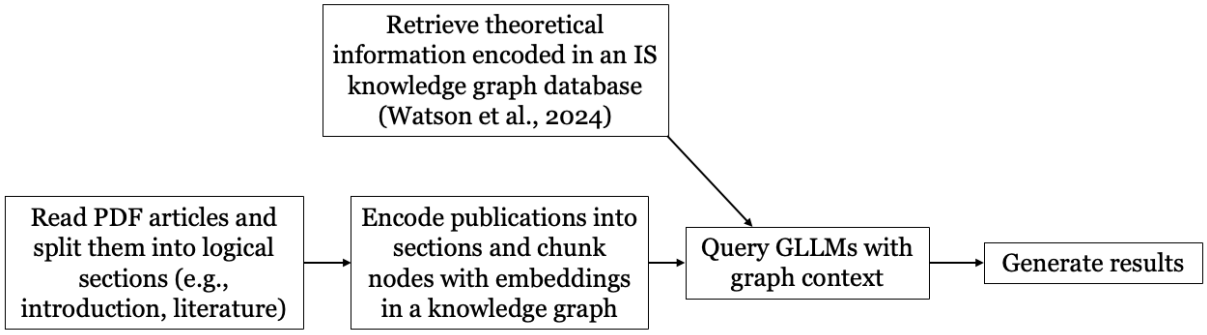


Figure 2. Design of the First Prototype of Our RAG-GLLM Literature Review Approach

Approach	Human	Non-retrieval GLLM (baseline)	RAG-GLLM
Extracted concepts	**Client participation, off-shore IS project success, information exchange activities, trust, differences in norms, project leader cultural values, differences in values**	Type of project, tacit knowledge, business domain expertise, technical development expertise, social embeddedness perspective, project characteristics, agency-related factors, relational exchange mechanism, **offshore IS project success**, multi-level research, outsourcing success, psychological contract perspective, supply chain logistics relationships	Strategic project, integration of tacit knowledge, business domain expertise, technical development expertise, social embeddedness perspective, **offshore IS project success**, project characteristics, agency-related factors, relational exchange mechanisms, **client participation**, **information exchange activities**, **trust**, *vendor-client work practices differences*, **leader-client representative cultural differences**, **project leader cultural values.**
Accuracy	100%	Precision=8%, recall=14%, F1=10%	Precision=43%, recall=93%, F1=59%
Note: bold concepts = correct extraction, italic concepts = semantically equivalent correct extraction			

Table 3. Comparison of Concept Extraction Performance of Different Approaches

This research contributes methodological and practical value to the IS discipline. Methodologically, we introduce a novel knowledge-based RAG-GLLM approach that enforces provenance, supports iterative synthesis, and scales across multiple levels in literature reviews. Practically, our approach addresses a pressing challenge faced by scholars in an era of exponential publication growth. By improving both efficiency and reliability, our design has the potential to reduce barriers to theory development and accelerate evidence-based knowledge discovery. In doing so, this research lays the groundwork for a new generation of AI-augmented scholarly practices that balance scalability with scientific rigour.

5 References

Abu-Salih, B., Alotaibi, S., Alanazi, A. L., Khurma, R. A., Al-Shboul, B., Khouri, A., and Aljaafari, M. 2025. "Using Large Language Models for Semantic Interoperability: A Systematic Literature Review," *ICT Express* (11:4), pp. 819-837.

Agarwal, S., Sahu, G., Puri, A., Laradji, I. H., Dvijotham, K. D., Stanley, J., Charlin, L., and Pal, C. 2024. "Litllm: A Toolkit for Scientific Literature Review," *arXiv preprint arXiv:2402.01788*.

Amabile, T. M., Conti, R., Coon, H., Lazenby, J., and Herron, M. 1996. "Assessing the Work Environment for Creativity," *Academy of Management Journal* (39:5), pp. 1154-1184.

An, S., Ma, Z., Lin, Z., Zheng, N., Lou, J.-G., and Chen, W. 2024. "Make Your Llm Fully Utilize the Context," *Advances in Neural Information Processing Systems* (37), pp. 62160-62188.

Arslan, M., Ghanem, H., Munawar, S., and Cruz, C. 2024. "A Survey on Rag with Llms," *Procedia Computer Science* (246), pp. 3781-3790.

Avgerou, C. 2019. "Contextual Explanation," *MIS Quarterly* (43:3), pp. 977-A918.

Babaei Giglou, H., D'Souza, J., and Auer, S. 2024. "Llms4synthesis: Leveraging Large Language Models for Scientific Synthesis," *Proceedings of the 24th ACM/IEEE Joint Conference on Digital Libraries*, pp. 1-12.

Baumeister, R. F., and Leary, M. R. 1997. "Writing Narrative Literature Reviews," *Review of General Psychology* (1:3), pp. 311-320.

Bernard, N., Sagawa Jr, Y., Bier, N., Lihoreau, T., Pazart, L., and Tannou, T. 2025. "Using Artificial Intelligence for Systematic Review: The Example of Elicit," *BMC Medical Research Methodology* (25:1), p. 75.

Bianchi, J., Hirt, J., Vogt, M., and Vetsch, J. 2025. "Data Extractions Using a Large Language Model (Elicit) and Human Reviewers in Randomized Controlled Trials: A Systematic Comparison," *Cochrane Evidence Synthesis and Methods* (3:4), p. e70033.

Chang, C., Wang, S., Zhang, J., Ge, J., and Li, L. 2024. "Llmscenario: Large Language Model Driven Scenario Generation," *IEEE Transactions on Systems, Man, and Cybernetics: Systems* (54:11), pp. 6581-6594.

de Kok, T. 2025. "Chatgpt for Textual Analysis? How to Use Generative Llms in Accounting Research," *Management Science* (Articles in Advance).

Delgado-Chaves, F. M., Jennings, M. J., Atalaia, A., Wolff, J., Horvath, R., Mamdouh, Z. M., Baumbach, J., and Baumbach, L. 2025. "Transforming Literature Screening: The Emerging Role of Large Language Models in Systematic Reviews," *Proceedings of the National Academy of Sciences* (122:2), p. e2411962122.

Dennstädt, F., Zink, J., Putora, P. M., Hastings, J., and Cihoric, N. 2024. "Title and Abstract Screening for Literature Reviews Using Large Language Models: An Exploratory Study in the Biomedical Domain," *Systematic Reviews* (13:1), p. 158.

Donthu, N., Kumar, S., Mukherjee, D., Pandey, N., and Lim, W. M. 2021. "How to Conduct a Bibliometric Analysis: An Overview and Guidelines," *Journal of business research* (133), pp. 285-296.

Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, H., and Wang, H. 2023. "Retrieval-Augmented Generation for Large Language Models: A Survey," *arXiv preprint arXiv:2312.10997* (2:1).

Gregor, S. 2006. "The Nature of Theory in Information Systems," *MIS Quarterly* (30:3), pp. 611-642.

Huang, Y., and Huang, J. 2024. "A Survey on Retrieval-Augmented Text Generation for Large Language Models," *arXiv preprint arXiv:2404.10981*.

Indulska, M., Hovorka, D. S., and Recker, J. 2012. "Quantitative Approaches to Content Analysis: Identifying Conceptual Drift across Publication Outlets," *European Journal of Information Systems* (21:1), pp. 49-69.

Jeyaraj, A., and Dwivedi, Y. K. 2020. "Meta-Analysis in Information Systems Research: Review and Recommendations," *International Journal of Information Management* (55), p. 102226.

Lála, J., O'Donoghue, O., Shtedritski, A., Cox, S., Rodriques, S. G., and White, A. D. 2023. "Paperqa: Retrieval-Augmented Generative Agent for Scientific Research," *arXiv preprint arXiv:2312.07559*.

Larsen, K. R., and Bong, C. H. 2016. "A Tool for Addressing Construct Identity in Literature Reviews and Meta-Analyses," *MIS Quarterly* (40:3), pp. 529-552.

Lee, S., Song, J., and Kim, Y. 2010. "An Empirical Comparison of Four Text Mining Methods," *Journal of Computer Information Systems* (51:1), pp. 1-10.

Levitt, H. M., Morrill, Z., Collins, K. M., and Rizo, J. L. 2021. "The Methodological Integrity of Critical Qualitative Research: Principles to Support Design and Research Review," *Journal of Counseling Psychology* (68:3), p. 357.

Rai, A., Maruping, L. M., and Venkatesh, V. 2009. "Offshore Information Systems Project Success: The Role of Social Embeddedness and Cultural Characteristics," *MIS Quarterly* (33:3), pp. 617-641.

Recker, J., Indulska, M., Green, P., Burton-Jones, A., and Weber, R. 2019. "Information Systems as Representations: A Review of the Theory and Evidence," *Journal of the Association for Information Systems* (20:6), pp. 735-786.

Sidorova, A., Evangelopoulos, N., Valacich, J. S., and Ramakrishnan, T. 2008. "Uncovering the Intellectual Core of the Information Systems Discipline," *MIS Quarterly*, pp. 467-482.

Taleb, I., Navaz, A. N., and Serhani, M. A. 2024. "Leveraging Large Language Models for Enhancing Literature-Based Discovery," *Big Data and Cognitive Computing* (8:11), pp. 1-28.

Tang, X., Duan, X., and Cai, Z. G. 2024. "Large Language Models for Automated Literature Review: An Evaluation of Reference Generation, Abstract Writing, and Review Composition," *arXiv preprint arXiv:2412.13612*.

Tong, S., Mao, K., Huang, Z., Zhao, Y., and Peng, K. 2024. "Automating Psychological Hypothesis Generation with Ai: When Large Language Models Meet Causal Graph," *Humanities and Social Sciences Communications* (11:1), pp. 1-14.

Tuczek, M., Degirmenci, K., Song, Y., Desouza, K. C., Breitner, M. H., and Watson, R. T. 2025. "Strategic Implications of Cognitive Computing in Is: Addressing Ai Fragmentation through Knowledge Similarity Transformation," *The Journal of Strategic Information Systems* (34:2), p. 101908.

Vindrola-Padros, C., and Johnson, G. A. 2020. "Rapid Techniques in Qualitative Research: A Critical Review of the Literature," *Qualitative health research* (30:10), pp. 1596-1604.

Wasserkrug, S., Boussioux, L., and Sun, W. 2024. "Combining Large Language Models and or/Ms to Make Smarter Decisions," in *Tutorials in Operations Research: Smarter Decisions for a Better World*. INFORMS, pp. 1-49.

Watson, R., Song, Y., Zhao, X., and Webster, J. 2024. "Extending the Foresight of Phillip Ein-Dor: Causal Knowledge Analytics," *Journal of the Association for Information Systems* (25:1), pp. 145-157.

Webster, J., and Watson, R. T. 2002. "Analyzing the Past to Prepare for the Future: Writing a Literature Review," *MIS Quarterly* (26:2), pp. Xiii-Xxiii.

Xu, D., Chen, W., Peng, W., Zhang, C., Xu, T., Zhao, X., Wu, X., Zheng, Y., Wang, Y., and Chen, E. 2024. "Large Language Models for Generative Information Extraction: A Survey," *Frontiers of Computer Science* (18:6), p. 186357.

Ye, X., Sun, R., Arik, S. Ö., and Pfister, T. 2023. "Effective Large Language Model Adaptation for Improved Grounding and Citation Generation," *arXiv preprint arXiv:2311.09533*.

Yoganarasimhan, H., and Iakovetskaia, I. 2024. "From Feeds to Inboxes: A Comparative Study of Polarization in Facebook and Email News Sharing," *Management Science* (70:9), pp. 6461-6472.

Zhou, Y., Liu, H., Srivastava, T., Mei, H., and Tan, C. 2024. "Hypothesis Generation with Large Language Models," *arXiv preprint arXiv:2404.04326*.